

The combinatorial invariance conjecture for KL polynomials

Grant Barkley
University of Michigan

October 8, 2026

joint with Christian Gaetz (Berkeley) and Thomas Lam (Michigan)

- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture
- 4 Diamond-generating sets
- 5 Richardson varieties and the Gabber–Joseph Conjecture
- 6 Increasing paths and frozen variables
- 7 Hypercube decompositions

Coxeter groups

A **Coxeter system** is a group W with generating set S subject to the following relations:

- $s^2 = e$ for all $s \in S$, and
- $(s_i s_j)^{m_{ij}} = e$ for $s_i \neq s_j \in S$ and $m_{ij} \in \{2, 3, \dots\} \cup \{\infty\}$.

Example

The following are Coxeter groups:

- The dihedral group $\langle s, t \mid s^2 = t^2 = (st)^n = e \rangle$.
- The symmetric group S_n , with $s_i = (i \ i+1)$, satisfying $(s_i s_j)^2 = e$ if $|i - j| > 1$ and $(s_i s_{i+1})^3 = e$.
- The Weyl group of any semisimple Lie group (or Kac–Moody group).
- The automorphism groups of the Platonic solids.

Length

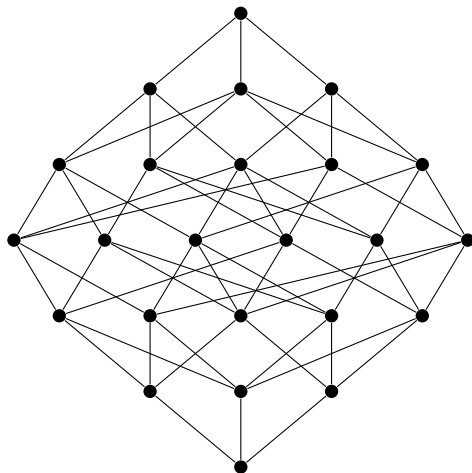
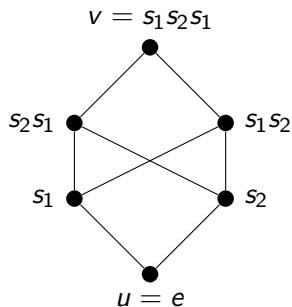
A **reduced word** for $w \in W$ is a minimum length expression $w = s_{i_1} \cdots s_{i_\ell}$. In this case $\ell = \ell(w)$ is called the **length** of w .

In the symmetric group S_n , the length of a permutation is the number of inversions it has:

$$\ell(231) = 2.$$

Bruhat order is the partial order $\leq$ on W which puts $u \leq v$ if there is a reduced word for u which is contained in a reduced word for v .

Bruhat order



Kazhdan–Lusztig polynomials

Associated to any two elements $u, v \in W$ is a polynomial $P_{u,v}(q)$:

$$P_{123,321} = 1$$

$$P_{1324,3412} = 1 + q$$

$$P_{21,12} = 0$$

This is the **Kazhdan–Lusztig polynomial** of u, v

$P_{u,v} \neq 0$ if and only if $u \leq v$ in Bruhat order

Kazhdan–Lusztig polynomials

The Kazhdan–Lusztig polynomials $P_{u,v}(q)$ for $u, v \in W$ are important numerical invariants in geometry, algebra, and representation theory.

Singularities of Schubert varieties

Representations of $\mathfrak{g}$

$$P_{u,v}(q)$$

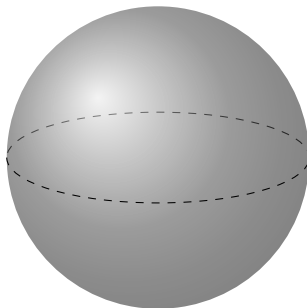
Hecke algebra of W

Motivation 1: Singularities of Schubert varieties

For each $v \in S_n$, there is a **Schubert variety** $\overline{X}_v$.

.

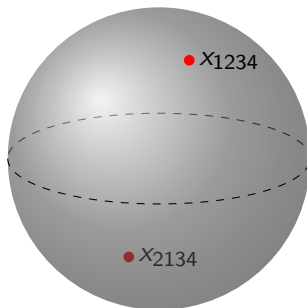
$$\overline{X}_{2134} = \mathbb{P}^1$$



Motivation 1: Singularities of Schubert varieties

For each $v \in S_n$, there is a **Schubert variety** $\overline{X}_v$.
 If $u \leq v$, then there is an associated point $x_u \in \overline{X}_v$.

$$\overline{X}_{2134} = \mathbb{P}^1$$

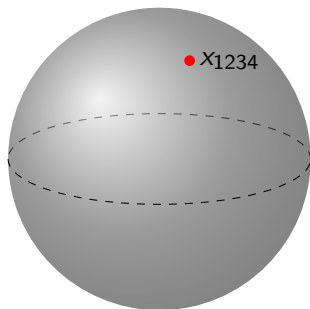


Motivation 1: Singularities of Schubert varieties

$P_{u,v}(q)$ describes the singularities of $\overline{X}_v$ near the point x_u

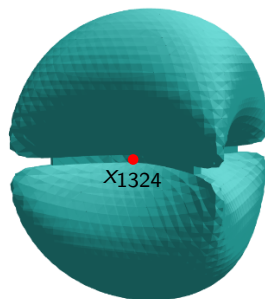
Example: $P_{u,v}(q) = 1$ if and only if $\overline{X}_v$ is smooth at x_u

$\overline{X}_{2134}$



$$P_{1234,2134} = 1$$

$\overline{X}_{3412}$



$$P_{1324,3412} = 1 + q$$

Motivation 2: Representations of $\mathfrak{gl}_n$

Let $\lambda = (\lambda_1, \dots, \lambda_n)$ and $\mu = (\mu_1, \dots, \mu_n)$ be sequences of n integers

Associated to λ and μ is a vector space V_λ^μ from representation theory:

“The μ -weight space of the irreducible $\mathfrak{gl}_n$ -representation of highest weight λ ”

The **character** of V_λ is $\text{ch}(V_\lambda) = \sum_{\mu} \dim(V_\lambda^\mu) \mathbf{x}^\mu$

If λ is a partition (decreasing sequence of non-negative integers), then $\text{ch}(V_\lambda)$ is a polynomial (a **Schur polynomial**)

Motivation 2: Representations of $\mathfrak{gl}_n$

$$\rho = (n, n-1, \dots, 2, 1)$$

Weyl character formula: if λ is a partition then

$$\text{ch}(V_\lambda) = \frac{\sum_{w \in S_n} (-1)^w \mathbf{x}^{w(\lambda+\rho)-\rho}}{\prod_{\alpha \in \Delta^+} (1 - \mathbf{x}^{-\alpha})}$$

The **Kazhdan–Lusztig conjecture**: if $v(\lambda + \rho)$ is a partition* then

$$\text{ch}(V_\lambda) = \frac{\sum_{w \in S_n} (-1)^w P_{ww_0, v w_0}(1) \mathbf{x}^{w(\lambda+\rho)-\rho}}{\prod_{\alpha \in \Delta^+} (1 - \mathbf{x}^{-\alpha})}$$

*: with all entries distinct

What is a KL polynomial?

What is a KL polynomial?

Motivation 1: the Poincaré polynomial of the stalk of an **intersection cohomology** complex on $\overline{X}_v$

$$P_{u,v}(q) = \sum_i q^i \dim H^{-\ell(u,v)+2i} \mathbf{IC}(\overline{X}_v)_{x_u}$$

Motivation 2: the Poincaré polynomial of the $R\mathrm{Hom}$ complex between Verma and simple $\mathfrak{g}$ -modules

$$P_{u,v}(q) = \sum_i q^i \dim \mathrm{Ext}^{\ell(u,v)-2i}(M_u, L_v)$$

Motivation 3: transition matrix for canonical basis of Hecke algebra

$$C_v = q^{-\frac{\ell(v)}{2}} \sum_{u \in S_n} P_{u,v}(q) T_u$$

What is a KL polynomial?

The **Kazhdan–Lusztig polynomials** are the unique family of polynomials $P_{u,v}(q) \in \mathbb{Z}[q]$ satisfying:

- (1) $P_{u,v}(q) = 0$ unless $u \leq v$.
- (2) $P_{u,u}(q) = 1$ for all u .
- (3) $\deg P_{u,v}(q) < \frac{1}{2}(\ell(v) - \ell(u))$ if $u < v$.
- (4) And, if $u < v$ then

$$q^{\ell(u,v)} P_{u,v} \left(\frac{1}{q} \right) - P_{u,v}(q) = \sum_{x \in (u,v]} R_{u,x}(q) P_{x,v}(q).$$

What is an R -polynomial?

One answer:

Associated to u and v is a **Richardson variety** $\mathcal{R}_{u,v}$. Then

$$R_{u,v}(q) = |\mathcal{R}_{u,v}(\mathbb{F}_q)|.$$

What is an R -polynomial?

The **R -polynomials** are the unique family of polynomials $R_{u,v}(q) \in \mathbb{Z}[q]$ satisfying:

- (1) $R_{u,v}(q) = 0$ unless $u \leq v$.
- (2) $R_{u,u}(q) = 1$ for all u .
- (3) If $us < u$ and $vs < v$:

$$R_{u,v}(q) = R_{us,vs}(q).$$

- (4) If $us > u$ and $vs < v$:

$$R_{u,v}(q) = qR_{us,vs}(q) + (q - 1)R_{u,vs}(q).$$

In summary...

In summary...

- You can compute $P_{u,v}$

In summary...

- You can compute $P_{u,v}$
- But, it's complicated

In summary...

- You can compute $P_{u,v}$
- But, it's complicated
- First you need to compute $R_{u,v}$

In summary...

- You can compute $P_{u,v}$
- But, it's complicated
- First you need to compute $R_{u,v}$
- And that's also complicated (big recursion)
- But the end result encodes important numerical invariants from different fields

Mysteries about Kazhdan–Lusztig polynomials

Kazhdan–Lusztig conjecture

Why is $P_{u,v}$ in representation theory?

Non-negativity conjecture

Why are the coefficients of $P_{u,v}$ always positive?

Combinatorial invariance conjecture

Why is $P_{u,v}$ determined by the Bruhat graph?

Mysteries about Kazhdan–Lusztig polynomials

Kazhdan–Lusztig conjecture

Solved in 1981

Beilinson–Bernstein, Brylinski–Kashiwara

Why is $P_{u,v}$ in representation theory?

Non-negativity conjecture

Why are the coefficients of $P_{u,v}$ always positive?

Combinatorial invariance conjecture

Why is $P_{u,v}$ determined by the Bruhat graph?

Mysteries about Kazhdan–Lusztig polynomials

Kazhdan–Lusztig conjecture

Solved in 1981

Beilinson–Bernstein, Brylinski–Kashiwara

Why is $P_{u,v}$ in representation theory?

Non-negativity conjecture

Solved in 2014

Elias–Williamson

Why are the coefficients of $P_{u,v}$ always positive?

Combinatorial invariance conjecture

Why is $P_{u,v}$ determined by the Bruhat graph?

Mysteries about Kazhdan–Lusztig polynomials

Kazhdan–Lusztig conjecture

Solved in 1981

Beilinson–Bernstein, Brylinski–Kashiwara

Why is $P_{u,v}$ in representation theory?

Non-negativity conjecture

Solved in 2014

Elias–Williamson

Why are the coefficients of $P_{u,v}$ always positive?

Combinatorial invariance conjecture

Solved?

Why is $P_{u,v}$ determined by the Bruhat graph?

- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph**
- 3 The Combinatorial Invariance Conjecture
- 4 Diamond-generating sets
- 5 Richardson varieties and the Gabber–Joseph Conjecture
- 6 Increasing paths and frozen variables
- 7 Hypercube decompositions

Reflections

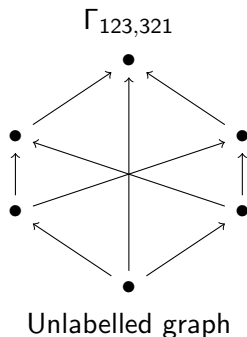
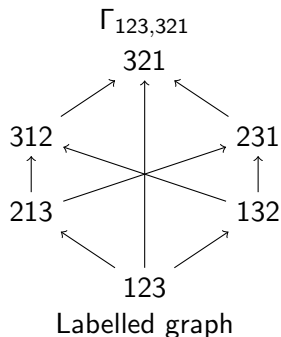
A **reflection** in W is an element of the form $ws w^{-1}$ for some $w \in W$ and $s \in S$.

In S_n , the reflections are exactly the transpositions.

Bruhat graphs

The **Bruhat graph** $\Gamma_{u,v}$ is a directed graph

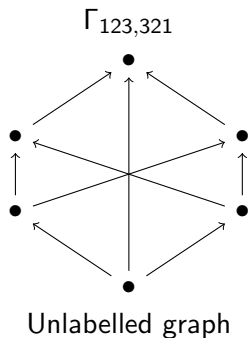
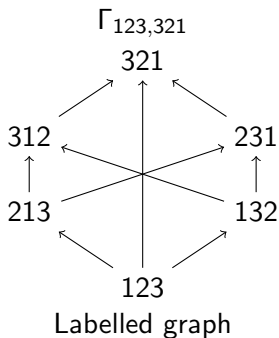
- Vertex set: Vertices are $x \in W$ so that $u \leq x$ and $x \leq v$
- Edges: There is an edge $x \rightarrow y$ if $\ell(x) < \ell(y)$ and we can transform x into y by multiplying by a reflection



Bruhat graphs

The **Bruhat graph** $\Gamma_{u,v}$ is a directed graph

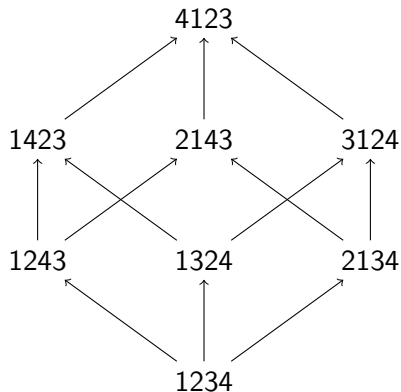
- Vertex set: Vertices are $x \in W$ so that $u \leq x$ and $x \leq v$
- Edges: There is an edge $x \rightarrow y$ if $\ell(x) < \ell(y)$ and we can transform x into y by multiplying by a reflection



Bruhat order is the transitive closure of the $\rightarrow$ relation

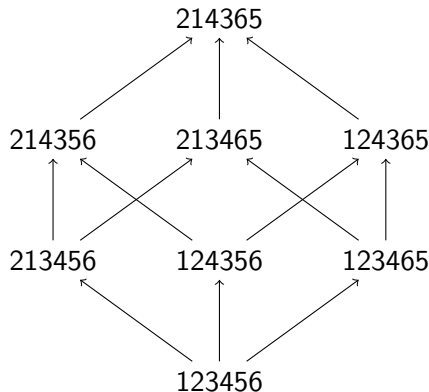
- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture**
- 4 Diamond-generating sets
- 5 Richardson varieties and the Gabber–Joseph Conjecture
- 6 Increasing paths and frozen variables
- 7 Hypercube decompositions

Bruhat graphs and polynomials



$$R_{1234,4123} = (q - 1)^3$$

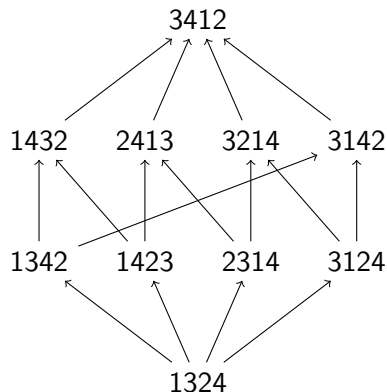
$$P_{1234,4123} = 1$$



$$R_{123456,214365} = (q - 1)^3$$

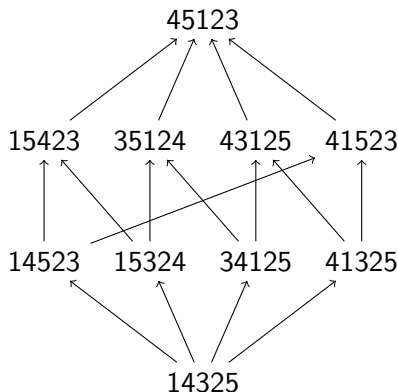
$$P_{123456,214365} = 1$$

Bruhat graphs and polynomials



$$R_{1324,3412} = (q - 1)^3$$

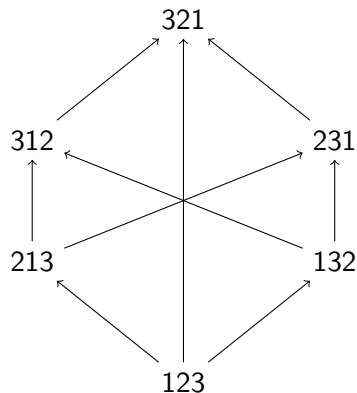
$$P_{1324,3412} = 1 + q$$



$$R_{14325,45123} = (q - 1)^3$$

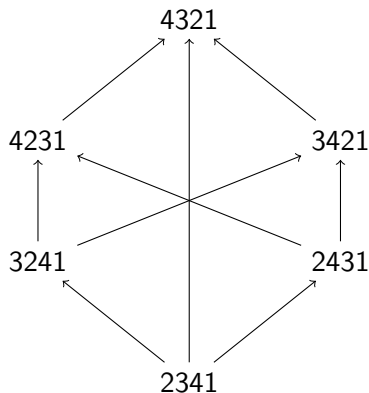
$$P_{14325,45123} = 1 + q$$

Bruhat graphs and polynomials



$$R_{123,321} = (q-1)^3 + q(q-1)$$

$$P_{123,321} = 1$$



$$R_{2341,4321} = (q-1)^3 + q(q-1)$$

$$P_{2341,4321} = 1$$

Combinatorial Invariance Conjecture

Lusztig and Dyer made the following surprising conjecture:

Conjecture (Lusztig 1980s; Dyer 1987)

If $\Gamma_{u,v}$ and $\Gamma_{u',v'}$ are isomorphic unlabelled directed graphs, then

$$P_{u,v}(q) = P_{u',v'}(q) \quad \text{and} \quad R_{u,v}(q) = R_{u',v'}(q).$$

- Jantzen 1979, Dyer 1987: true when $P_{u,v}(q) = 1$.
- Brenti 1994: true when all edges of $\Gamma_{u,v}$ have length 1.
- Dyer 1987, Incitti 2006 & 2007: true if $|W| < \infty$ and $\ell(u, v) \leq 8$.
- Brenti, du Cloux, Delanoy, Brenti–Caselli–Marietti 2003–2006: true when $u = u' = e = 123 \cdots n$.

Combinatorial Invariance Conjecture

Most of these results in fact prove:

Strong Combinatorial Invariance Conjecture

Given $\Gamma_{u,v}$ as an unlabelled directed graph, there is an algorithm for computing $P_{u,v}(q)$ from $\Gamma_{u,v}$.

- Jantzen 1979, Dyer 1987: true when $P_{u,v}(q) = 1$.
- Brenti 1994: true when all edges of $\Gamma_{u,v}$ have length 1.
- Dyer 1987, Incitti 2006 & 2007: true if $|W| < \infty$ and $\ell(u, v) \leq 8$.
- Brenti, du Cloux, Delanoy, Brenti–Caselli–Marietti 2003-2006: true when $u = u' = e = 123 \cdots n$.

Examples of combinatorial invariance

Theorem (Brenti 1994)

The polynomial $R_{u,v}(q)$ factors as $(q-1)^{\ell(u,v)}$ if and only if all edges of $\Gamma_{u,v}$ have length 1.

General fact:

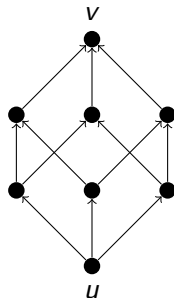
If we know $R_{x,y}$ for every $x, y \in \Gamma_{u,v}$, then we can also compute $P_{u,v}$

Examples of combinatorial invariance



$$R_{u,v}(q) = q - 1$$
$$P_{u,v}(q) = 1$$

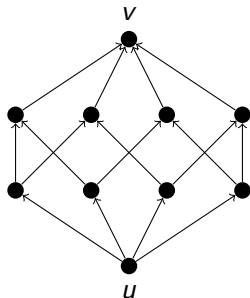
Examples of combinatorial invariance



$$R_{u,v}(q) = (q - 1)^3$$

$$P_{u,v}(q) = 1$$

Examples of combinatorial invariance



$$R_{u,v}(q) = (q - 1)^3$$

$$P_{u,v}(q) = 1 + q$$

$\tilde{R}$ -polynomials

Definition (Dyer)

There is a unique family of polynomials $\tilde{R}_{uv}(q) \in \mathbb{N}[q]$ such that

$$q^{\frac{\ell(v)-\ell(u)}{2}} \tilde{R}_{uv}(q^{\frac{1}{2}} - q^{-\frac{1}{2}}) = R_{uv}(q).$$

The CIC can equivalently be stated for any of the P -, R -, or $\tilde{R}$ -polynomials.

Conjecture (Equivalent form of Combinatorial Invariance Conjecture)

If $[u, v] \cong [u', v']$ as posets, then $R_{uv}(q) = R_{u'v'}(q)$ and $\tilde{R}_{uv}(q) = \tilde{R}_{u'v'}(q)$.

CIC for coefficients

Theorem (Barkley–G.–Lam, 2026; Finite types ADE due to Patimo, 2021)

Let $u, v \in W$ and $u', v' \in W'$ and suppose that $[u, v] \cong [u', v']$. Then:

- $[q]P_{uv} = [q]P_{u'v'}$
- $[q]R_{uv} = [q]R_{u'v'}$ and $[q^{\ell-1}]R_{uv} = [q^{\ell-1}]R_{u'v'}$
- $[q^{\ell-2}]\tilde{R}_{uv} = [q^{\ell-2}]\tilde{R}_{u'v'}$

	q^0	q^1	q^2	$\dots$	$q^{\ell(u,v)-2}$	$q^{\ell(u,v)-1}$	$q^{\ell(u,v)}$
$P_{u,v}$	KL79	BGL26	?	?	—	—	—
$R_{u,v}$	KL79	BGL26	?	?	?	BGL26	KL79
$\tilde{R}_{u,v}$	KL79	Dyer87	Dyer87	?	BGL26	—	KL79

Table: Coefficients of the P -, R -, and $\tilde{R}$ -polynomials which are known to be combinatorially invariant for all Coxeter groups.

CIC for short intervals

	q^0	q^1	q^2	$\dots$	$q^{\ell(u,v)-2}$	$q^{\ell(u,v)-1}$	$q^{\ell(u,v)}$
$P_{u,v}$	KL79	BGL26	?	?	—	—	—
$R_{u,v}$	KL79	BGL26	?	?	?	BGL26	KL79
$\tilde{R}_{u,v}$	KL79	Dyer87	Dyer87	?	BGL26	—	KL79

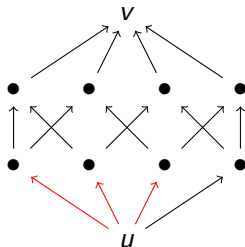
Corollary (Barkley-G.-Lam, 2026)

The CIC holds for intervals with $\ell(u, v) \leq 6$.

- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture
- 4 Diamond-generating sets**
- 5 Richardson varieties and the Gabber–Joseph Conjecture
- 6 Increasing paths and frozen variables
- 7 Hypercube decompositions

Diamond-generating sets

Let E_{uv} denote the set of cover relations of $[u, v]$. A subset $X \subset E_{uv}$ is called **diamond-closed** if, whenever it contains two adjacent edges from some diamond, it also contains the other two. Write $\overline{X}$ for the smallest diamond-closed set containing X . Patimo defined g_{uv} to be the smallest size of X with $\overline{X} = E_{uv}$.



$$d_{uv} = g_{uv}$$

For $u \leq v$ define $d_{uv} \in \mathbb{N}$ by

$$R_{u,v} = q^{\ell(u,v)} - d_{uv}q^{\ell(u,v)-1} + \dots$$

We also have:

$$\begin{aligned} [q]P_{uv} &= \#\{\text{coatoms of } [u, v]\} - d_{uv} \\ [q^{\ell(u,v)-2}] \tilde{R}_{uv} &= \ell(u, v) - d_{uv}. \end{aligned}$$

Theorem (Barkley–G.–Lam, 2026; Finite types ADE due to Patimo, 2021)

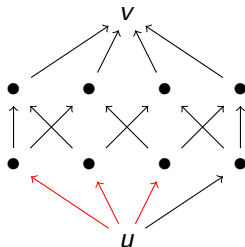
Let $u \leq v$ be elements in any Coxeter group W . Then $d_{uv} = g_{uv}$.

Proof idea.

It follows from work of Dyer that $d_{uv} \leq g_{uv}$. We construct an explicit diamond-generating set F_{uv} of size d_{uv} , motivated by the cluster structure on $\mathcal{R}_{uv}$. □

$$d_{uv} = g_{uv}$$

We have $g_{uv} = d_{uv} = 3$ for this interval.



- $P_{uv} = 1 + (4 - 3)q$
- $\tilde{R}_{uv} = (3 - 3)q + q^3$
- $R_{uv} = -1 + 3q - 3q^2 + q^3$

- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture
- 4 Diamond-generating sets
- 5 Richardson varieties and the Gabber–Joseph Conjecture
- 6 Increasing paths and frozen variables
- 7 Hypercube decompositions

Richardson varieties

Let W be the Weyl group of a semisimple Lie group G with Borel subgroup B , and $u \leq v \in W$. The **(open) Richardson variety** is

$$\mathcal{R}_{uv} = BvB/B \cap B^-uB/B \subset G/B.$$

It is known that for all prime powers q we have

$$\#(\mathcal{R}_{uv})_{\mathbb{F}_q} = R_{uv}(q).$$

Verma modules and the Gabber–Joseph conjecture

Let $\mathfrak{g}$ be a complex semisimple Lie algebra. Write M_ν for the **Verma module** with highest weight $\nu\lambda - \rho$, for any anti-dominant regular integral weight λ .

These are infinite-dimensional $\mathfrak{g}$ -modules, with submodules corresponding to Bruhat order on W .

Gabber and Joseph defined the R' polynomial by:

$$R'_{uv}(q) = \sum_{i=0}^{\ell(u,v)} (-1)^{\ell(u,v)-i} q^i \dim \operatorname{Ext}_{\mathcal{O}}^i(M_u, M_v).$$

Gabber–Joseph for $\text{Ext}^{\ell-1}$

Conjecture (Gabber–Joseph Conjecture, 1981)

Let $u \leq v \in W$. Then $R_{uv} = R'_{uv}$.

- It follows from [Verma 1968] that $[q^0]R'_{u,v} = [q^0]R_{u,v}$.

We prove the conjecture for the coefficient of $q^{\ell(u,v)-1}$.

Theorem (Barkley–G.–Lam, 2026)

For $u \leq v \in W$ we have

$$\dim \text{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v) = d_{u,v}.$$

Consequently $\dim \text{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v)$ is a combinatorial invariant.

Gabber–Joseph for $\text{Ext}^{\ell-1}$

Conjecture (Gabber–Joseph Conjecture, 1981)

Let $u \leq v \in W$. Then $R_{uv} = R'_{uv}$.

- It follows from [Verma 1968] that $[q^0]R'_{u,v} = [q^0]R_{u,v}$.
- [Carlin 1986] showed that $[q^{\ell(u,v)}]R'_{u,v} = [q^{\ell(u,v)}]R_{u,v}$.

We prove the conjecture for the coefficient of $q^{\ell(u,v)-1}$.

Theorem (Barkley–G.–Lam, 2026)

For $u \leq v \in W$ we have

$$\dim \text{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v) = d_{u,v}.$$

Consequently $\dim \text{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v)$ is a combinatorial invariant.

Gabber–Joseph for $\text{Ext}^{\ell-1}$

Conjecture (Gabber–Joseph Conjecture, 1981)

Let $u \leq v \in W$. Then $R_{uv} = R'_{uv}$.

- It follows from [Verma 1968] that $[q^0]R'_{u,v} = [q^0]R_{u,v}$.
- [Carlin 1986] showed that $[q^{\ell(u,v)}]R'_{u,v} = [q^{\ell(u,v)}]R_{u,v}$.
- A *counterexample* to the full conjecture was found by Boe in 1992.

We prove the conjecture for the coefficient of $q^{\ell(u,v)-1}$.

Theorem (Barkley–G.–Lam, 2026)

For $u \leq v \in W$ we have

$$\dim \text{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v) = d_{u,v}.$$

Consequently $\dim \text{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v)$ is a combinatorial invariant.

Cohomology of open Richardsons

Theorem (Barkley–G.–Lam, 2026)

For $u \leq v \in W$ we have

$$\dim \operatorname{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v) = d_{u,v}.$$

Consequently $\dim \operatorname{Ext}_{\mathcal{O}}^{\ell(u,v)-1}(M_u, M_v)$ is a combinatorial invariant.

Really what we prove is:

Proposition

The mixed Hodge spaces $H^{1,(0,0)}(\mathcal{R}_{u,v}, \mathbb{C})$ and $H^{2,(1,1)}(\mathcal{R}_{u,v}, \mathbb{C})$ vanish.
Hence

$$d_{u,v} = \dim H^1(\mathcal{R}_{u,v}, \mathbb{C}).$$

- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture
- 4 Diamond-generating sets
- 5 Richardson varieties and the Gabber–Joseph Conjecture
- 6 Increasing paths and frozen variables**
- 7 Hypercube decompositions

$\tilde{R}$ -polynomials and reflection orders

The reflections T are in bijection with the positive **roots** $\Phi^+ \subset \mathbb{R}\{\alpha_s \mid s \in S\}$.

Definition (Dyer)

A **reflection order** is a total order $\prec$ on T so that for any triple $\alpha, \beta, \gamma \in \Phi^+$ with $\gamma \in \mathbb{R}_{>0}\alpha + \mathbb{R}_{>0}\beta$ we have either

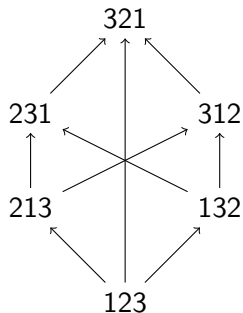
$$\alpha \prec \gamma \prec \beta \quad \text{or} \quad \beta \prec \gamma \prec \alpha.$$

Theorem (Dyer 1987)

Fix a reflection order $\prec$. The coefficient of q^k in $\tilde{R}_{uv}$ is the number of **increasing paths** in Γ from u to v of length k .

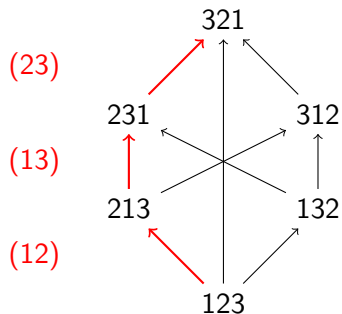
$\tilde{R}$ -polynomial example

$$(1, 2) \prec (1, 3) \prec (2, 3)$$



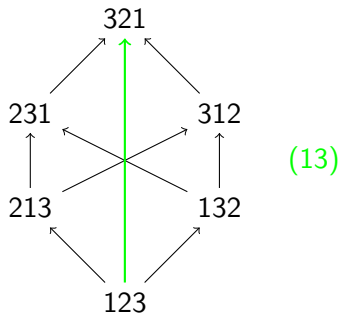
$\tilde{R}$ -polynomial example

$$(1, 2) \prec (1, 3) \prec (2, 3)$$



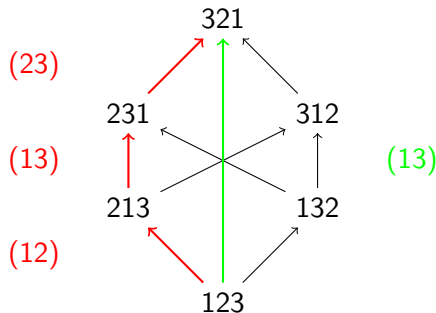
$\tilde{R}$ -polynomial example

$$(1, 2) \prec (1, 3) \prec (2, 3)$$



$\tilde{R}$ -polynomial example

$$(1, 2) \prec (1, 3) \prec (2, 3)$$



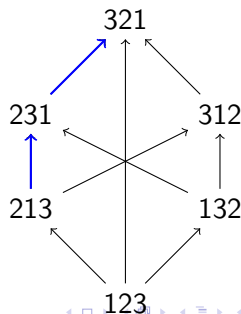
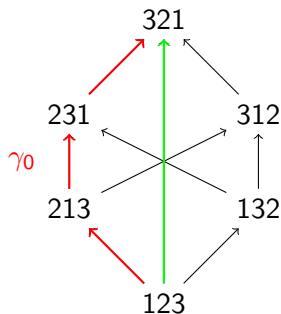
$$\tilde{R}_{123,321} = q^3 + q^1$$

So we expect $g_{uv} = 2$

Divergences of increasing paths

Theorem (Barkley–G.–Lam, 2026)

Fix a reflection order $\prec$ such that the inversions of v are a prefix of $\prec$. Let γ_0 be the unique increasing maximal chain from u to v . Then at most one increasing $(\ell - 2)$ -path diverges from γ_0 at each point. Let F_{uv} be the set of edges of γ_0 with no such divergence. Then $\#F_{uv} = d_{uv}$ and F_{uv} is a diamond-generating set.



Application to cluster structure

For finite Weyl groups, the coordinate ring $\mathbb{C}[\mathcal{R}_{uv}]$ is known to be a **cluster algebra** (Casals–Gorsky–Gorsky–Le–Shen–Simental & Galashin–Lam–Sherman–Bennett–Speyer). Each extended cluster has size $\ell(u, v) = \dim(\mathcal{R}_{uv})$ and is composed of **frozen** and **mutable** cluster variables.

Corollary

Under the natural bijection between increasing paths and distinguished subexpressions, F_{uv} maps to the frozen variables of this cluster structure. Consequently, there are d_{uv} and $\ell(u, v) - d_{uv}$ frozen and mutable variables in each cluster, and these quantities are combinatorially invariant.

- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture
- 4 Diamond-generating sets
- 5 Richardson varieties and the Gabber–Joseph Conjecture
- 6 Increasing paths and frozen variables
- 7 Hypercube decompositions**

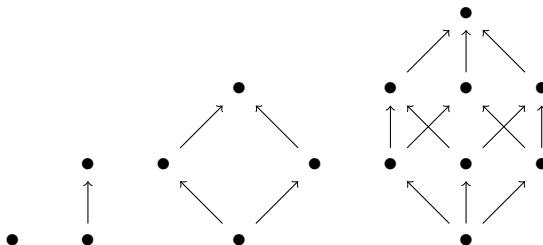
Machine learning Kazhdan–Lusztig polynomials

Blundell–Buesing–Davies–Veličković (from Google DeepMind) and Geordie Williamson trained a neural network to guess Kazhdan–Lusztig polynomials, given the poset.

- They asked the neural net which substructures of $\Gamma_{u,v}$ affected its answer a lot.
- It told* them about **hypercube decompositions**.
- They conjectured a recurrence for the $P_{u,v}(q)$ using hypercube decompositions which would imply the Combinatorial Invariance Conjecture.

Hypercubes

A **hypercube** is a directed graph that is the poset diagram of a Boolean lattice



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

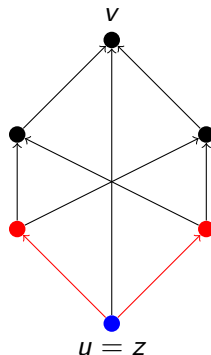
A **hypercube decomposition** of $\Gamma_{u,v}$ consists of:

- a subgraph $I = \Gamma_{u,z}$ and,
- for each $x \in I$ and $y \notin I$, the set of hypercubes $\mathcal{H}_x^y$ with bottom vertex x , top vertex y , and no edges in I

The hypercubes must satisfy:

- An existence condition
- A uniqueness condition: No two hypercubes have the same bottom edges

Non-example



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

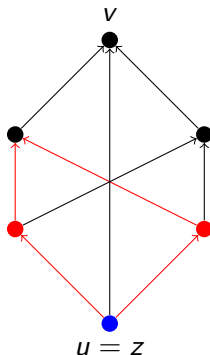
A **hypercube decomposition** of $\Gamma_{u,v}$ consists of:

- a subgraph $I = \Gamma_{u,z}$ and,
- for each $x \in I$ and $y \notin I$, the set of hypercubes $\mathcal{H}_x^y$ with bottom vertex x , top vertex y , and no edges in I

The hypercubes must satisfy:

- An existence condition
- A uniqueness condition: No two hypercubes have the same bottom edges

Non-example



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

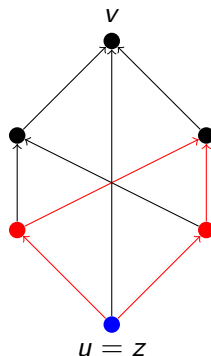
A **hypercube decomposition** of $\Gamma_{u,v}$ consists of:

- a subgraph $I = \Gamma_{u,z}$ and,
- for each $x \in I$ and $y \notin I$, the set of hypercubes $\mathcal{H}_x^y$ with bottom vertex x , top vertex y , and no edges in I

The hypercubes must satisfy:

- An existence condition
- A uniqueness condition: No two hypercubes have the same bottom edges

Non-example



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

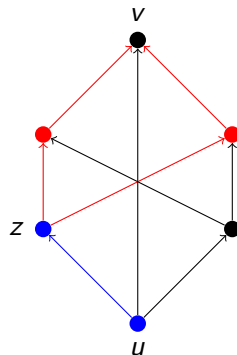
A **hypercube decomposition** of $\Gamma_{u,v}$ consists of:

- a subgraph $I = \Gamma_{u,z}$ and,
- for each $x \in I$ and $y \notin I$, the set of hypercubes $\mathcal{H}_x^y$ with bottom vertex x , top vertex y , and no edges in I

The hypercubes must satisfy:

- An existence condition
- A uniqueness condition: No two hypercubes have the same bottom edges

Example



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

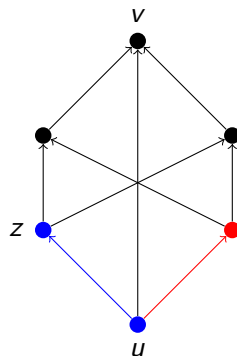
A **hypercube decomposition** of $\Gamma_{u,v}$ consists of:

- a subgraph $I = \Gamma_{u,z}$ and,
- for each $x \in I$ and $y \notin I$, the set of hypercubes $\mathcal{H}_x^y$ with bottom vertex x , top vertex y , and no edges in I

The hypercubes must satisfy:

- An existence condition
- A uniqueness condition: No two hypercubes have the same bottom edges

Example



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

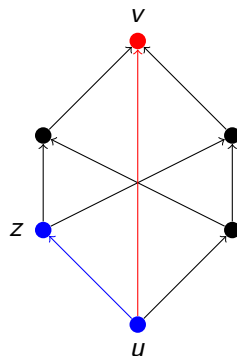
A **hypercube decomposition** of $\Gamma_{u,v}$ consists of:

- a subgraph $I = \Gamma_{u,z}$ and,
- for each $x \in I$ and $y \notin I$, the set of hypercubes $\mathcal{H}_x^y$ with bottom vertex x , top vertex y , and no edges in I

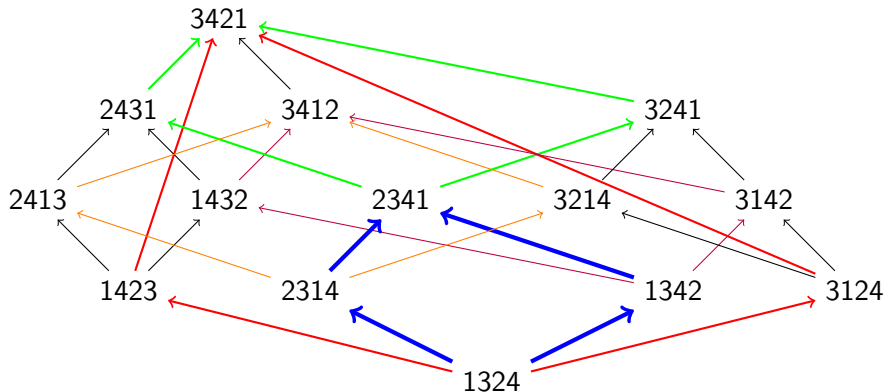
The hypercubes must satisfy:

- An existence condition
- A uniqueness condition: No two hypercubes have the same bottom edges

Example



A bigger example



The BBDVW conjecture

Let $\text{rk}(H)$ denote the rank of a hypercube H , so that $|H| = 2^{\text{rk}(H)}$

Conjecture (Blundell–Buesing–Davies–Veličković–Williamson 2022)

Let I be any hypercube decomposition of $\Gamma_{u,v}$. Then

$$q^{\ell(u,v)} P_{u,v} \left(\frac{1}{q} \right) - P_{u,v} = \sum_{x \in I \setminus \{u\}} R_{u,x} P_{x,v} - q^{\ell(u,v)} \sum_{\substack{y \notin I \\ H \in \mathcal{H}_u^y}} (q^{-1} - 1)^{\text{rk}(H)} q^{-|\Lambda(H)|} P_{y,v} \left(\frac{1}{q} \right).$$

Has a natural geometric interpretation in this form

Proved by BBDVW for **standard hypercube decompositions**

Theorem

Theorem (B.–Gaetz 2025)

The BBDVW recursion holds for any hypercube decomposition of a Bruhat graph with the form $\Gamma_{e,v}$.

Theorem

Theorem (B.–Gaetz 2025)

The BBDVW recursion holds for any hypercube decomposition of a Bruhat graph with the form $\Gamma_{e,v}$.

We also have generalizations to intervals with $[q]P_{uw_0, vw_0} = 0$, under some hypotheses

Theorem

Theorem (B.–Gaetz 2025)

The BBDVW recursion holds for any hypercube decomposition of a Bruhat graph with the form $\Gamma_{e,v}$.

We also have generalizations to intervals with $[q]P_{uw_0, vw_0} = 0$, under some hypotheses

These are the only cases of the BBDVW conjecture proven so far

A simplification: $\tilde{R}$ -polynomials

The BBDVW formula involves R -polynomials and P -polynomials and is very complicated. Is there a simpler way?

Definition (Brenti)

There is a unique family of polynomials $\tilde{R}_{u,v}(q) \in \mathbb{N}[q]$ such that

$$q^{\frac{\ell(v)-\ell(u)}{2}} \tilde{R}_{u,v}(q^{\frac{1}{2}} - q^{-\frac{1}{2}}) = R_{u,v}(q).$$

$$\begin{aligned} \text{Examples: } R_{u,v} &= (q-1)^5 & \rightsquigarrow & \tilde{R}_{u,v} = q^5 \\ R_{u,v} &= (q-1)^3 + q(q-1) & \rightsquigarrow & \tilde{R}_{u,v} = q^3 + q \end{aligned}$$

Equivalent form of Combinatorial Invariance Conjecture

If $\Gamma_{u,v} \cong \Gamma_{u',v'}$, then $\tilde{R}_{u,v}(q) = \tilde{R}_{u',v'}(q)$.

A conjecture

Conjecture (B.–Gaetz 2023, B.–Gaetz 2025)

Let $(I, \mathcal{H})$ be any hypercube decomposition of $\Gamma_{u,v}$ which satisfies a “numerical criterion”. Then

$$\sum_{\substack{y \notin I \\ H \in \mathcal{H}_u^y}} (-q)^{\text{rk}(H)} \tilde{R}_{y,v} = 0$$

and

$$\tilde{R}_{u,v} = \sum_{x \in I} \sum_{H \in \mathcal{H}_x^v} q^{\text{rk}(H)} \tilde{R}_{u,x}.$$

A conjecture

Conjecture (B.–Gaetz 2023, B.–Gaetz 2025)

Let $(I, \mathcal{H})$ be any hypercube decomposition of $\Gamma_{u,v}$ which satisfies a “numerical criterion”. Then

$$\sum_{\substack{y \notin I \\ H \in \mathcal{H}_u^y}} (-q)^{\text{rk}(H)} \widetilde{R}_{y,v} = 0$$

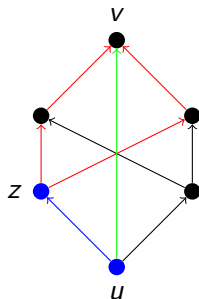
and

$$\widetilde{R}_{u,v} = \sum_{x \in I} \sum_{H \in \mathcal{H}_x^v} q^{\text{rk}(H)} \widetilde{R}_{u,x}.$$

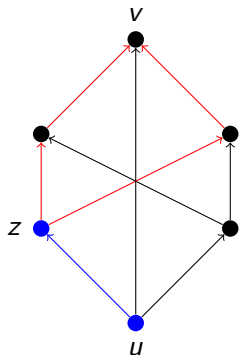
Either formula would imply the combinatorial invariance conjecture!

Example

$$\begin{aligned}\tilde{R}_{u,v} &= \sum_{x \in I} \sum_{H \in \mathcal{H}_x^v} q^{\text{rk}(H)} \tilde{R}_{u,x} = \textcolor{green}{q} \tilde{R}_{u,u} + \textcolor{red}{q}^2 \tilde{R}_{u,z} \\ &= \textcolor{green}{q} \cdot \textcolor{blue}{1} + \textcolor{red}{q}^2 \cdot \textcolor{blue}{q} = q + q^3 = \tilde{R}_{u,v}.\end{aligned}$$



Thank you!



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

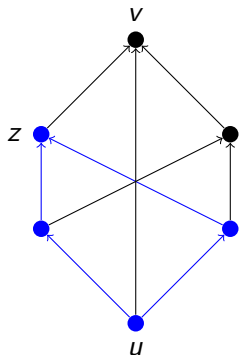
A subgraph $I = \Gamma_{u,z} \subset \Gamma_{u,v}$ forms a **hypercube decomposition** of $\Gamma_{u,v}$ if:

- (1) I is **diamond-closed** inside $\Gamma_{u,v}$.
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.

A is an **antichain** if there is no path between any two elements of A

Non-example of hypercube decomposition

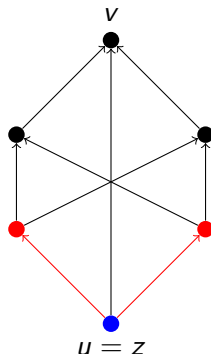
(1) I is **diamond-closed** inside $[u, v]$.



$I = \Gamma_{uz}$ not diamond-closed

Non-example of hypercube decomposition

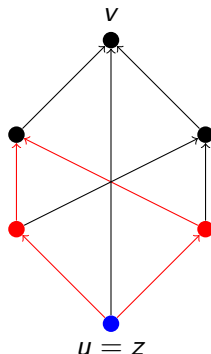
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.



$I = \Gamma_{uz}$ is diamond-closed but hypercube is not unique

Non-example of hypercube decomposition

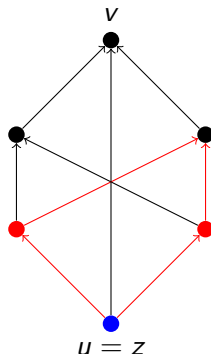
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.



$I = \Gamma_{uz}$ is diamond-closed but hypercube is not unique

Non-example of hypercube decomposition

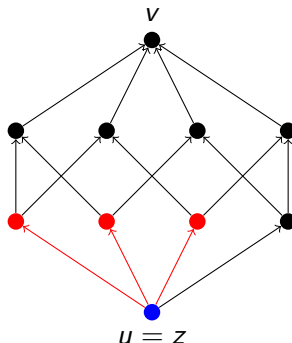
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.



$I = \Gamma_{uz}$ is diamond-closed but hypercube is not unique

Non-example of hypercube decomposition

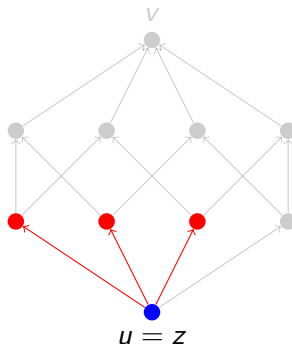
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.



$I = \Gamma_{u,z}$ is diamond-closed but hypercube does not exist

Non-example of hypercube decomposition

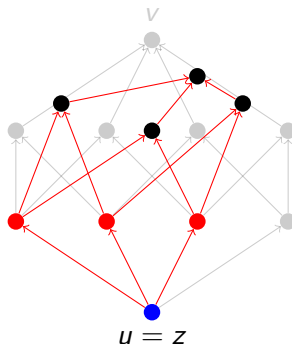
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.



$I = \Gamma_{u,z}$ is diamond-closed but hypercube does not exist

Non-example of hypercube decomposition

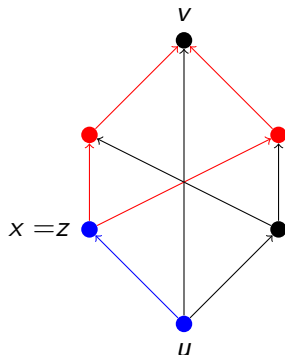
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.



$I = \Gamma_{u,z}$ is diamond-closed but hypercube does not exist

Example of hypercube decomposition

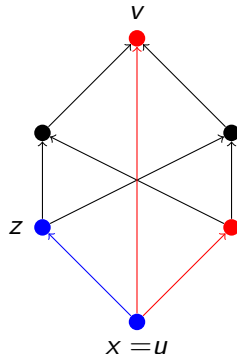
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a **unique hypercube** in $\Gamma_{u,v}$.



$I = \Gamma_{u,z}$ is diamond-closed and hypercubes are unique

Example of hypercube decomposition

- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a **unique hypercube** in $\Gamma_{u,v}$.

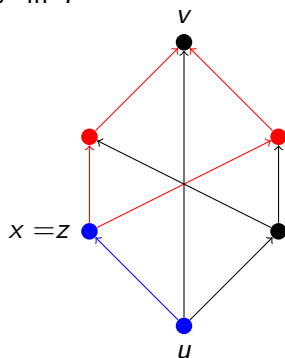


$I = \Gamma_{u,z}$ is diamond-closed and hypercubes are unique

The hypercube map

Let $\mathcal{I}$ is a hypercube decomposition and $Y \subseteq \mathcal{Y}_x$

- $\theta_x(Y) :=$ the top of the hypercube spanned by the antichain of maximal elements¹ in Y



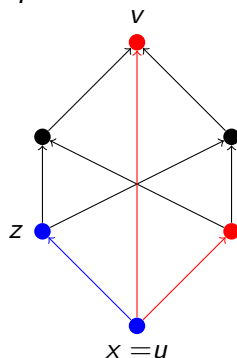
$$\theta_x(\mathcal{Y}_x) = v$$

¹elements of Y that do not have a path to another element of Y

The hypercube map

Let $\mathcal{I}$ is a hypercube decomposition and $Y \subseteq \mathcal{Y}_x$

- $\theta_x(Y) :=$ the top of the hypercube spanned by the antichain of maximal elements¹ in Y



$$\theta_x(\mathcal{Y}_x) = v$$

¹elements of Y that do not have a path to another element of Y

The numerical criterion

Let I be a hypercube decomposition of $[u, v]$.

Definition

We say I satisfies the **numerical criterion** (at u) if, for all $Y \in \mathcal{A}_u$,

$$\ell(u, \theta_u(Y)) = 2 |\{x \in \mathcal{Y}_u \mid \exists y \in Y, x \leq y\}| - |Y|.$$

Lemma

If I satisfies the numerical criterion, then

$$\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A), v} = 0$$

implies the BBDVW conjecture for I .

The numerical criterion

Let I be a hypercube decomposition of $[u, v]$.

Definition

We say I satisfies the **numerical criterion** (at u) if, for all $Y \in \mathcal{A}_u$,

$$\ell(u, \theta_u(Y)) = 2 |\{x \in \mathcal{Y}_u \mid \exists y \in Y, x \leq y\}| - |Y|.$$

Lemma

If I satisfies the numerical criterion, then

$$\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A), v} = 0$$

implies the BBDVW conjecture for I .

Remark: sometimes $\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A), v}$ is nonzero!

The theorem

Theorem (B.–Gaetz 24+)

If I is a hypercube decomposition of an interval of the form $[e, v]$ (a **lower interval**), then I satisfies the numerical criterion and

$$\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \widetilde{R}_{\theta(A), v} = 0.$$

In particular, the BBDVW conjecture is true for lower intervals.

The theorems

Theorem (B.–Gaetz 24+)

If I is a hypercube decomposition of an interval of the form $[e, v]$ (a **lower interval**), then I satisfies the numerical criterion and

$$\sum_{A \in \mathcal{A}_e} (-1)^{|A|} q^{|A|} \widetilde{R}_{\theta(A), v} = 0.$$

In particular, the BBDVW conjecture is true for lower intervals.

Theorem (B.–Gaetz 23)

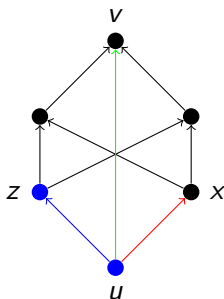
If I is a hypercube decomposition of a lower interval $[e, v]$, then

$$\widetilde{R}_{ev} = \sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A) = v}} q^{|A|} \widetilde{R}_{ex}.$$

Example of BBDVW

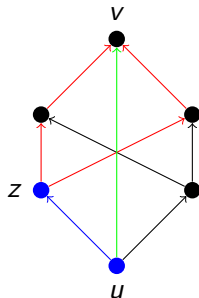
$$\sum_{A \in \mathcal{A}_e} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A), v} = \tilde{R}_{u, v} - \textcolor{green}{q} \tilde{R}_{vv} - \textcolor{red}{q} \tilde{R}_{xv}(q)$$

$$= q + q^3 - \textcolor{green}{q} \cdot 1 - \textcolor{red}{q} \cdot q^2 = 0.$$



Example of BG23

$$\begin{aligned}
 \tilde{R}_{u,v} &= \sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A) = v}} q^{|A|} \tilde{R}_{u,x} = \textcolor{green}{q} \tilde{R}_{u,u} + \textcolor{red}{q}^2 \tilde{R}_{u,z} \\
 &= \textcolor{green}{q} \cdot \textcolor{blue}{1} + \textcolor{red}{q}^2 \cdot \textcolor{blue}{q} = q + q^3 = \tilde{R}_{u,v}.
 \end{aligned}$$



The conjecture

Conjecture

Let I be any hypercube decomposition of $\Gamma_{u,v}$ which satisfies the numerical criterion. Then

$$\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A),v} = 0$$

and

$$\tilde{R}_{u,v} = \sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A)=v}} q^{|A|} \tilde{R}_{u,x}.$$

The proof

$\tilde{R}$ -polynomials and reflection orders

Definition

A **reflection order** is a total order $\prec$ on transpositions so that for all $1 \leq a \leq b \leq c \leq n$,

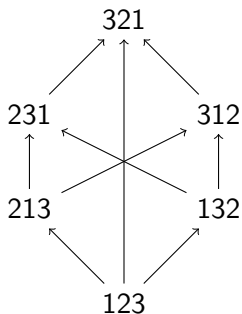
$$(ab) \prec (ac) \prec (bc) \quad \text{or} \quad (bc) \prec (ac) \prec (ab).$$

Theorem (Dyer 1987)

Fix a reflection order $\prec$. The coefficient of q^k in $\tilde{R}_{u,v}$ is the number of **increasing paths** from u to v of length k .

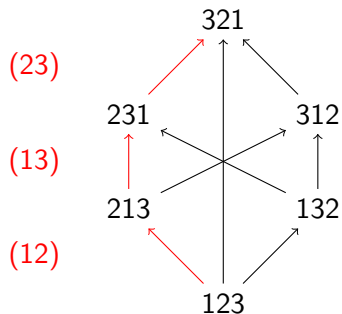
$\tilde{R}$ -polynomial example

$$(12) \prec (13) \prec (23)$$



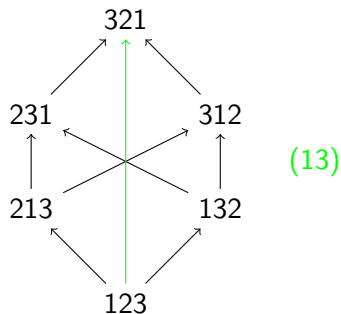
$\tilde{R}$ -polynomial example

$$(12) \prec (13) \prec (23)$$



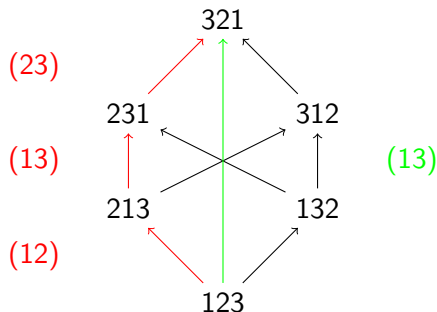
$\tilde{R}$ -polynomial example

$$(12) \prec (13) \prec (23)$$



$\tilde{R}$ -polynomial example

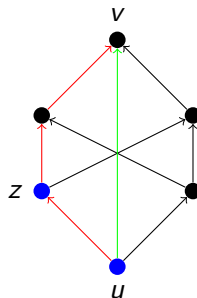
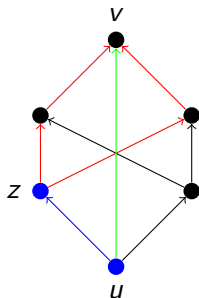
$$(12) \prec (13) \prec (23)$$



$$\tilde{R}_{123,321} = q^3 + q^1$$

Proof idea

$$\sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A) = v}} q^{|A|} \tilde{R}_{u,x} = \tilde{R}_{u,v},$$



The lemma

A rearrangement

BBDVW conjecture equivalent to

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} \sum_{y \in [u, v] \setminus I} R_{\theta(Y), y} P_{yv} = 0.$$

What if already

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} R_{\theta(Y), y} = 0?$$

A rearrangement

BBDVW conjecture equivalent to

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} \sum_{y \in [u, v] \setminus I} R_{\theta(Y), y} P_{yv} = 0.$$

What if already

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} R_{\theta(Y), y} = 0?$$

Let's simplify even more.

Simplifying further

Lemma

Assume that I is a hypercube decomposition of $[u, v]$ satisfying the numerical criterion. Then

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} R_{\theta(Y), y} = 0$$

if and only if

$$\sum_{Y \in \mathcal{A}_u} (-1)^{|Y|} q^{|Y|} \tilde{R}_{\theta(Y), y} = 0.$$